By the defn. of transpose, the ith row of B^T is the ith column of B.

$\therefore b'_{ik} = b_{kj}$, where $1 \leq k \leq n$.

By the defn. of transpose, the ith column of A^T is the ith row of A.

$\therefore a'_{ki} = a_{ik}$, where $1 \leq k \leq n$

$$\sum_{k=1}^{n} b'_{ik} \cdot a'_{ki} = \sum_{k=1}^{n} b_{kj} \cdot a_{ik} = \sum_{k=1}^{n} a_{ik} \cdot b_{kj} = (i,j)\text{th entry of } AB$$
$$= (j,i)\text{th entry of } (AB)^T$$

$\boxed{\therefore (AB)^T = B^T A^T}$ $\square$